

A Note on Defect States in Discrete Schrödinger Chains: Uniform Upper Bounds, Monotone Bracketing, and Localization Envelopes

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Abstract

We study a single on-site defect in a finite one-dimensional tight-binding / mass-spring chain. Our toy model is a row of identical sites (masses, optical waveguides, or atomic orbitals) coupled only to their nearest neighbours; one site is made slightly different, and this creates a localized “defect mode” whose frequency (or energy) lies outside the usual band of extended modes. Mathematically, this is encoded by a real-symmetric tridiagonal Toeplitz matrix $T_N(a, b)$ and a rank-one diagonal perturbation

$$H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top \text{ at site } k.$$

This paper proves three main results. First, we recast the defect eigenvalue $\lambda(\alpha, k)$ as the unique root of a scalar secular equation $f(\lambda) = 1 + \alpha m_k(\lambda) = 0$, where $m_k(\lambda)$ is a resolvent entry of $T_N(a, b)$, and show that this root lies outside the finite-chain band $[E_{\min}^{(N)}, E_{\max}^{(N)}]$. Second, we give simple Weyl/Gershgorin-type upper bounds for $\lambda(\alpha, k)$ and use them to build a monotone bisection bracket anchored just above $E_{\max}^{(N)}$. Third, whenever $\lambda(\alpha, k)$ lies strictly outside the surrogate band $[a - 2|b|, a + 2|b|]$, we obtain an explicit geometric envelope that controls the localization of the normalized defect eigenvector. The emphasis throughout is on a self-contained linear-algebra treatment of a standard physics model based on resolvents, rank-one updates, and localization estimates.

I. Introduction

Consider a line of identical masses connected by identical springs, or equally a row of identical optical waveguides or quantum wells arranged in a straight line. In such systems, the normal modes form a continuous-looking “band” of allowed frequencies or energies, and each mode is spread out across the entire chain. If we change the mass at a single site, or the refractive index of a single waveguide, we create a defect that can pin a new vibration or bound state whose amplitude is strongly concentrated near that site and whose frequency lies outside the band, as in standard discrete Schrödinger impurity models and scattering setups [3, 5].

After standard nondimensionalization, these examples reduce to an eigenvalue problem for a real-symmetric tridiagonal matrix. We model the unperturbed chain by a Toeplitz matrix $T_N(a, b)$ with constant diagonal and off-diagonal entries, and we model the defect by adding a rank-one diagonal per-

turbation $\alpha e_k e_k^\top$ at site k , giving the defect Hamiltonian $H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top$. Here N is the number of sites, a encodes the on-site term, b the nearest-neighbour coupling, and α the strength of the defect; this is the standard finite-chain discrete Schrödinger/tight-binding model with a single on-site impurity [3, 6]. The eigenvalues of $T_N(a, b)$ form a discrete “band” contained strictly in the surrogate interval $[a - 2|b|, a + 2|b|]$, while a sufficiently strong defect can create an additional outlier eigenvalue $\lambda(\alpha, k)$ with a strongly localized eigenvector. In solid-state physics this picture appears when modelling an impurity atom inside a crystal, and in photonics it describes a defect cavity in a coupled-resonator optical waveguide that can trap light at a prescribed frequency [5].

This paper addresses three basic questions that connect the physical picture to linear algebra: *(i)* when does a defect eigenvalue exist, and is it unique? *(ii)* how large can it be, in terms of N , a , b , and α , and how can we compute it reliably? *(iii)* how

rapidly does the associated defect eigenvector decay away from the defect site, and how does this depend on k (bulk versus near-edge defects)? Our first main result reformulates the defect eigenvalue as the unique root of a scalar equation involving a single resolvent entry of $T_N(a, b)$. The second result gives simple Weyl/Gershgorin-type upper bounds for this root and turns them into a practical monotone bisection scheme. The third result shows that, whenever the defect eigenvalue lies outside the surrogate band, the defect mode decays at least geometrically away from the defect site with an explicit envelope that matches numerical experiments.

Throughout, we use only real-symmetric matrices, spectral decompositions, resolvents, rank-one updates, and three-term recurrences; terms such as “Birman–Schwinger”, “Green’s function”, and “Combes–Thomas” are mentioned only as standard physics labels for these tools. The underlying model and rank-one-defect analysis are standard in the theory of discrete Schrödinger operators, but they are presented here in a concise form that makes the connection between matrix methods and physical impurity problems much clearer. Section II collects the preliminaries, Section III presents the main structural results, Section IV illustrates them numerically, Section V summarizes the computational methods, and the final sections discuss limitations and further directions.

II. Preliminaries

Standing assumptions. We fix the data

$$\begin{aligned} N \in \mathbb{N}, \quad a \in \mathbb{R}, \quad b \in \mathbb{R} \setminus \{0\}, \\ k \in \{1, \dots, N\}, \quad \alpha \in \mathbb{R} \setminus \{0\}, \end{aligned}$$

and work throughout with real-symmetric matrices. We write $T_N(a, b)$ (or simply T_N when a, b are fixed) for the unperturbed Toeplitz chain and $H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top$ for the defect Hamiltonian. Physically, N is the number of sites in the chain, k is the location of the defect, and α measures how much we have changed the local parameter (mass, refractive index, or on-site potential) at that site.

Toeplitz chain and defect Hamiltonian. The unperturbed chain is represented by the tridiagonal

Toeplitz matrix

$$T_N(a, b) = \begin{pmatrix} a & b & & & 0 \\ b & a & b & & \\ & b & a & \ddots & \\ & & \ddots & \ddots & b \\ 0 & & & b & a \end{pmatrix},$$

which can be viewed as a discrete Laplacian plus a constant shift. The defect Hamiltonian is obtained by changing only the k -th diagonal entry,

$$H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top,$$

so that all couplings remain nearest-neighbour and only one site is special. In physical language $H(\alpha, k)$ is a tight-binding Hamiltonian with a single impurity.

Unperturbed eigenvalues and finite-chain band edges. The eigenvalues of $T_N(a, b)$ are explicitly

$$\begin{aligned} \lambda_j^{(0)}(a, b, N) &= a + 2b \cos\left(\frac{j\pi}{N+1}\right), \\ j &= 1, \dots, N, \end{aligned}$$

so the finite-chain band edges are

$$\begin{aligned} E_{\max}^{(N)} &= a + 2|b| \cos\left(\frac{\pi}{N+1}\right) < a + 2|b|, \\ E_{\min}^{(N)} &= a - 2|b| \cos\left(\frac{\pi}{N+1}\right) > a - 2|b|. \end{aligned}$$

Thus the spectrum of $T_N(a, b)$ lies in the closed interval $[E_{\min}^{(N)}, E_{\max}^{(N)}]$, which itself is strictly contained in the surrogate $[a - 2|b|, a + 2|b|]$; this cosine band structure is standard for constant-coefficient Jacobi/Toeplitz matrices [4]. In the physics examples above this interval is the finite-chain analogue of the continuous energy band for an infinite periodic medium. Whenever the defect Hamiltonian $H(\alpha, k)$ has an eigenvalue lying outside $[E_{\min}^{(N)}, E_{\max}^{(N)}]$, we denote it by $\lambda(\alpha, k)$ and call it the *defect eigenvalue*, and we write $v^{(\alpha, k)}$ for a corresponding normalized eigenvector. Throughout the rest of the paper we regard N, a, b as fixed background parameters and view $\lambda(\alpha, k) = \lambda(\alpha, k; N, a, b)$ as a scalar function of the defect strength α and the defect site k ; Theorem 1 will show that, for each fixed k and each sign of α , there is at most one such out-of-band eigenvalue and that it is unique whenever it exists.

Characteristic polynomials and three-term recurrence. For each $n \geq 0$ we set

$$p_n(\lambda) = \det(T_n(a, b) - \lambda I_n),$$

with the convention $p_0(\lambda) = 1$. A standard expansion along the last row of $T_n(a, b) - \lambda I_n$ shows that the polynomials p_n satisfy the three-term recurrence

$$p_0(\lambda) = 1,$$

$$p_1(\lambda) = a - \lambda,$$

$$p_n(\lambda) = (a - \lambda)p_{n-1}(\lambda) - b^2 p_{n-2}(\lambda), \quad n \geq 2.$$

This is a second-order linear recurrence with constant coefficients, and its closed-form solution can be expressed in terms of Chebyshev polynomials (in particular those of the second kind) [2, 4].

Example 2.1 (A 4×4 Toeplitz chain). Take $a = 0$, $b = 1$, and $N = 4$. Then $T_4(0, 1)$ has characteristic polynomials

$$p_0(\lambda) = 1,$$

$$p_1(\lambda) = -\lambda,$$

$$p_2(\lambda) = \lambda^2 - 1,$$

$$p_3(\lambda) = -\lambda^3 + 2\lambda,$$

$$p_4(\lambda) = \lambda^4 - 3\lambda^2 + 1,$$

obtained directly from the recurrence. The eigenvalues of $T_4(0, 1)$ are the roots of $p_4(\lambda)$, and numerically one finds $\lambda \approx \pm 1.618$ and $\lambda \approx \pm 0.618$, all contained in the band $[-2, 2] = [a - 2|b|, a + 2|b|]$.

Resolvent entry and secular equation. For λ away from the spectrum of $T_N(a, b)$ we write

$$(T_N(a, b) - \lambda I_N)^{-1}$$

for the resolvent. Its diagonal entries are often called the discrete Green's function. We single out the (k, k) -entry

$$m_k(\lambda) := [(T_N(a, b) - \lambda I_N)^{-1}]_{kk},$$

which can be expressed in terms of the characteristic polynomials as

$$m_k(\lambda) = e_k^\top (T_N(a, b) - \lambda I_N)^{-1} e_k = \frac{p_{k-1}(\lambda) p_{N-k}(\lambda)}{p_N(\lambda)},$$

a standard Green's-function identity for Jacobi/Toeplitz matrices [4, 6]. Formally, $m_k(\lambda)$ measures the response at site k to a unit impulse applied at the same site at “energy” λ .

Example 2.2 (A small chain with a defect).

Let $a = 0$, $b = 1$, $N = 4$, $k = 2$, and $\alpha = 3$. From the previous example we have

$$m_2(\lambda) = \frac{p_1(\lambda) p_2(\lambda)}{p_4(\lambda)} = \frac{-\lambda(\lambda^2 - 1)}{\lambda^4 - 3\lambda^2 + 1},$$

so the secular equation $f(\lambda) = 1 + 3m_2(\lambda) = 0$ can be evaluated numerically for $\lambda > 2$. For example, at $\lambda = 2.5$ one finds

$$m_2(2.5) \approx -0.616,$$

$$f(2.5) = 1 + 3m_2(2.5) \approx -0.85 < 0,$$

which shows that the defect eigenvalue must lie to the right of 2.5. More generally, one finds $f(2.2) < 0$ and $f(3.0) > 0$, so there is a root $\lambda(\alpha, k) \in (2.2, 3.0)$ with $\lambda(\alpha, k) > E_{\max}^{(4)}$. This simple calculation foreshadows the general existence-and-uniqueness result and the bracketing algorithm proved later, and it illustrates how a single diagonal defect can create a localized mode above the band in a concrete matrix of size 4×4 .

Linear-algebra viewpoint and tools. From this point on we view the problem purely through linear algebra. The Toeplitz matrix $T_N(a, b)$ is real-symmetric with simple spectrum, so it admits an orthonormal eigenbasis and a spectral decomposition. The resolvent $(T_N(a, b) - \lambda I_N)^{-1}$ is well-defined off the spectrum and its diagonal entries $m_k(\lambda)$ play the role of discrete Green's functions. The rank-one perturbation $\alpha e_k e_k^\top$ leads to the scalar secular equation $1 + \alpha m_k(\lambda) = 0$, so the defect eigenvalue problem for $H(\alpha, k)$ is reduced to understanding a single rational function of λ . All of the later results are derived from this setup using only spectral decompositions, three-term recurrences, and standard inequalities for eigenvalues of Hermitian matrices; the labels “Birman–Schwinger”, “Green's function”, and “Combes–Thomas” correspond to the more general operator-theoretic frameworks in [1, 5, 7].

III. Main Results

Throughout this section we assume that $T_N(a, b)$ has simple spectrum with

$$E_{\min}^{(N)} = E_1 < E_2 < \cdots < E_N = E_{\max}^{(N)},$$

which holds for the tridiagonal Toeplitz matrices in our examples. For $\alpha > 0$, if $H(\alpha, k)$ has an eigenvalue on the right half-line $(E_{\max}^{(N)}, \infty)$ we denote it by $\lambda(\alpha, k)$ and call it the *right-hand* defect eigenvalue; for $\alpha < 0$, any eigenvalue of $H(\alpha, k)$ on the left half-line $(-\infty, E_{\min}^{(N)})$ is denoted by the same symbol and is called the *left-hand* defect eigenvalue. Theorem 1 below shows that, for each fixed k and sign of α , there is at most one such eigenvalue and that, whenever it exists, it is unique. Recall that $f(\lambda) = 1 + \alpha m_k(\lambda)$ is the scalar secular function built from the resolvent entry $m_k(\lambda)$.

Theorem 1 (Existence and uniqueness of the defect eigenvalue). *Fix $N \geq 2$ and a site $1 \leq k \leq N$, and assume that $T_N(a, b)$ has simple eigenvalues $E_1 < E_2 < \dots < E_N$.*

- (a) *For $\alpha > 0$ the function $f(\lambda) = 1 + \alpha m_k(\lambda)$ is strictly increasing on (E_N, ∞) , with*

$$\lim_{\lambda \downarrow E_N} f(\lambda) = -\infty, \quad \lim_{\lambda \rightarrow \infty} f(\lambda) = 1.$$

Hence there exists a unique root

$$\lambda(\alpha, k) \in (E_N, \infty)$$

and this root is an eigenvalue of $H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top$.

- (b) *For $\alpha < 0$ the analogous statement holds on $(-\infty, E_1)$: $f(\lambda)$ is strictly increasing there with $\lim_{\lambda \rightarrow -\infty} f(\lambda) = 1$ and $\lim_{\lambda \uparrow E_1} f(\lambda) = +\infty$, so there is a unique root $\lambda(\alpha, k) \in (-\infty, E_1)$, which is again an eigenvalue of $H(\alpha, k)$.*

In particular, there is at most one defect eigenvalue outside the finite band $[E_{\min}^{(N)}, E_{\max}^{(N)}]$ on each side.

Sketch of proof. Write the spectral decomposition

$$T_N(a, b) = \sum_{j=1}^N E_j w^{(j)} (w^{(j)})^\top,$$

where the eigenvectors $w^{(j)}$ form an orthonormal basis. For $\lambda \notin \{E_j\}$ one obtains the standard resolvent expansion

$$m_k(\lambda) = [(T_N(a, b) - \lambda I)^{-1}]_{kk} = \sum_{j=1}^N \frac{|w_k^{(j)}|^2}{E_j - \lambda}.$$

Differentiating termwise on each interval between eigenvalues gives

$$m'_k(\lambda) = \sum_{j=1}^N \frac{|w_k^{(j)}|^2}{(E_j - \lambda)^2} > 0,$$

so $m_k(\lambda)$ and hence $f(\lambda)$ are strictly increasing on each such interval. On the rightmost interval (E_N, ∞) the term with $j = N$ dominates as $\lambda \downarrow E_N$, and we have $m_k(\lambda) \rightarrow -\infty$ and thus $f(\lambda) \rightarrow -\infty$. As $\lambda \rightarrow \infty$ each fraction tends to 0, so $m_k(\lambda) \rightarrow 0$ and $f(\lambda) \rightarrow 1$. This proves part (a); part (b) is analogous on $(-\infty, E_1)$.

Finally, the rank-one perturbation identity or the cofactor formula for a diagonal update shows that zeros of $f(\lambda)$ are in one-to-one correspondence with eigenvalues of $H(\alpha, k)$ lying outside $[E_{\min}^{(N)}, E_{\max}^{(N)}]$. A general functional-analytic treatment of this monotonicity and the associated interlacing for Hermitian rank-one perturbations can be found in, for example, [1]. $\square$

Remark 1 (Defect eigenvalue notation and parameter dependence). For each fixed chain length N , background parameters a, b , defect site k , and sign of α , Theorem 1 shows that there is at most one eigenvalue of $H(\alpha, k)$ lying outside the finite band $[E_{\min}^{(N)}, E_{\max}^{(N)}]$, and that when it exists it is unique. From now on, $\lambda(\alpha, k) = \lambda(\alpha, k; N, a, b)$ always denotes this unique out-of-band eigenvalue, and $v^{(\alpha, k)}$ denotes a corresponding normalized eigenvector. Standard rank-one perturbation theory implies that, for each fixed k , the map $\alpha \mapsto \lambda(\alpha, k)$ is monotone in α ; in particular, for $\alpha > 0$ the right-hand defect eigenvalue lies in $(E_{\max}^{(N)}, \infty)$ and increases with α , while for $\alpha < 0$ the left-hand defect eigenvalue lies in $(-\infty, E_{\min}^{(N)})$ and decreases with α .

Theorem 2 (Weyl/Gershgorin upper bounds and bisection bracket). *Let $\lambda(\alpha, k)$ be the defect eigenvalue from Theorem 1 for $\alpha > 0$.*

Define

$$U_{\text{Weyl}}(\alpha) = E_{\max}^{(N)} + \alpha, \quad U_{\text{Gersh}}(\alpha) = a + \alpha + 2|b|,$$

and

$$U_{\text{bound}}(\alpha) := \min\{U_{\text{Weyl}}(\alpha), U_{\text{Gersh}}(\alpha)\}.$$

Then

$$\lambda(\alpha, k) \leq U_{\text{bound}}(\alpha),$$

and, for any $\varepsilon \in (0, 1)$ with $E_{\text{max}}^{(N)} + \varepsilon < U_{\text{bound}}(\alpha)$, the interval

$$[L_0, U_0] := [E_{\text{max}}^{(N)} + \varepsilon, U_{\text{bound}}(\alpha)]$$

satisfies $f(L_0) < 0 < f(U_0)$ and is a valid initial bracket for a monotone bisection scheme converging to $\lambda(\alpha, k)$.

Sketch of proof. The matrix $H(\alpha, k)$ can be written as a rank-one update

$$H(\alpha, k) = T_N(a, b) + \alpha e_k e_k^\top.$$

By the variational characterization of the largest eigenvalue,

$$\begin{aligned} \lambda_{\text{max}}(H(\alpha, k)) &= \max_{\|x\|=1} x^\top H(\alpha, k) x \\ &\leq \max_{\|x\|=1} x^\top T_N(a, b) x + \alpha \\ &= E_{\text{max}}^{(N)} + \alpha, \end{aligned}$$

which is the standard Weyl-type bound and yields

$$U_{\text{Weyl}}(\alpha).$$

For Gershgorin, each row of $H(\alpha, k)$ has diagonal entry $a + \alpha$ when $j = k$ and a otherwise, with at most two off-diagonal entries of magnitude $|b|$. Hence all eigenvalues lie in the union of discs

$$\{z \in \mathbb{C} : |z - a| \leq 2|b|\} \cup \{z \in \mathbb{C} : |z - (a + \alpha)| \leq 2|b|\},$$

so, in particular, $\lambda(\alpha, k) \leq a + \alpha + 2|b| = U_{\text{Gersh}}(\alpha)$. Taking the minimum of the two bounds gives $\lambda(\alpha, k) \leq U_{\text{bound}}(\alpha)$.

By Theorem 1, $f(\lambda)$ is strictly increasing on $(E_{\text{max}}^{(N)}, \infty)$ with $f(\lambda) \rightarrow -\infty$ as $\lambda \downarrow E_{\text{max}}^{(N)}$ and $f(\lambda) \rightarrow 1 > 0$ as $\lambda \rightarrow U_{\text{bound}}(\alpha)$. Thus there exists $\varepsilon > 0$ small enough that $f(E_{\text{max}}^{(N)} + \varepsilon) < 0$, and with this choice one has $f(L_0) < 0 < f(U_0)$ as claimed. $\square$

Bisection algorithm. Given $\alpha > 0$ and the bracket $[L_0, U_0] = [E_{\text{max}}^{(N)} + \varepsilon, U_{\text{bound}}(\alpha)]$ from Theorem 2, we locate $\lambda(\alpha, k)$ by the following monotone bisection procedure:

1. Set $n = 0$ and choose a tolerance $\tau > 0$.

2. At step n , define the midpoint $M_n = (L_n + U_n)/2$ and evaluate $f(M_n)$.
3. If $f(M_n) \leq 0$, set $L_{n+1} = M_n$, $U_{n+1} = U_n$; otherwise set $L_{n+1} = L_n$, $U_{n+1} = M_n$.
4. Stop when $U_n - L_n < \tau$ and report any point in $[L_n, U_n]$ as an approximation of $\lambda(\alpha, k)$.

Strict monotonicity of f on $(E_{\text{max}}^{(N)}, \infty)$ (Theorem 1) guarantees that the true defect eigenvalue always remains in $[L_n, U_n]$ and that $L_n \uparrow \lambda(\alpha, k)$, $U_n \downarrow \lambda(\alpha, k)$.

Example 3.1 (Two bracketing steps). Take $a = 0$, $b = 1$, $N = 200$, $k = 100$, and $\alpha = 5$, as in Fig. 1. One computes $E_{\text{max}}^{(N)} \approx 1.9998$ and hence $U_{\text{bound}}(5) \approx 7.0$. Choosing $\varepsilon = 10^{-2}$ gives the initial bracket $[L_0, U_0] \approx [2.01, 7.0]$. A direct evaluation shows $f(L_0) < 0 < f(U_0)$; the midpoint $M_0 \approx 4.50$ becomes the new left or right endpoint depending on the sign of $f(M_0)$. After one or two such steps the interval has already shrunk significantly, and continuing the process converges rapidly to the defect eigenvalue plotted in Fig. 1.

Theorem 3 (Geometric localization outside the finite band). *Assume that the defect eigenvalue $\lambda(\alpha, k)$ lies strictly outside the finite band $[E_{\text{min}}^{(N)}, E_{\text{max}}^{(N)}]$, and let $v^{(\alpha, k)}$ be a normalized eigenvector of $H(\alpha, k)$ for $\lambda(\alpha, k)$. Then there exist constants $C > 0$ and $\rho \in (0, 1)$, depending on a , b , and $\lambda(\alpha, k)$ but not on j , such that*

$$|v_j^{(\alpha, k)}| \leq C \rho^{|j-k|} \quad \text{for all } j = 1, \dots, N.$$

In particular, if $r_{\pm}$ are the roots of

$$br^2 + (a - \lambda)r + b = 0, \quad (1)$$

then for $\lambda \notin [a - 2|b|, a + 2|b|]$ one has $|r_-| < 1 < |r_+|$ and we may take $\rho = |r_-|$ and, with $d = \min\{k - 1, N - k\}$,

$$|v_j^{(\alpha, k)}| \leq \frac{|r_-|^{|j-k|}}{\sqrt{(1 - |r_-|^2)(1 - |r_-|^{2d+2})}} \quad \text{for all } j. \quad (2)$$

Sketch of proof. Away from the defect site k , the eigenvector components satisfy the homogeneous three-term recurrence

$$bv_{j+1} + (a - \lambda)v_j + bv_{j-1} = 0, \quad j \neq k.$$

Looking for solutions of the form $v_j \propto r^{j-k}$ leads to the quadratic equation (1). When λ lies outside the band $[a - 2|b|, a + 2|b|]$, a standard discriminant computation shows that there are two real roots $r_{\pm}$ with $|r_-| < 1 < |r_+|$ and $r_+ r_- = 1$. Any solution of the recurrence can therefore be written as a linear combination of the two geometric sequences r_-^{j-k} and r_+^{j-k} on each side of the defect. Rewriting the recurrence as

$$v_{j+1} = -\frac{a-\lambda}{b} v_j - v_{j-1}$$

and taking absolute values gives the explicit inequality

$$|v_{j+1}| \leq \frac{|a-\lambda|}{|b|} |v_j| + |v_{j-1}|,$$

so once $|v_j|$ and $|v_{j-1}|$ are controlled by a multiple of $|r_-|^{|j-k|}$, the same bound propagates to $|v_{j+1}|$.

The Dirichlet boundary conditions at $j = 0$ and $j = N + 1$ together with normalization of $v^{(\alpha,k)}$ force the coefficients of the growing mode r_+^{j-k} to be small. A geometric-series estimate on each half-chain then yields a bound of the form $|v_j^{(\alpha,k)}| \leq C |r_-|^{|j-k|}$, where the constant C depends only on a , b , and $\lambda(\alpha, k)$ and is independent of j . A more careful book-keeping of the two-sided normalization on the finite chain gives the explicit denominator in (2), which improves the constant near the boundaries through the factor involving $d = \min\{k - 1, N - k\}$. This envelope is the geometric curve overlaid on the numerical eigenvector in Fig. 2. Sharper versions of such localization bounds are available in the literature on discrete Schrödinger operators; see, for example, [7, 6, 3]. $\square$

IV. Numerical validation (two figures)

We now illustrate Theorems 1, 2, and 3 with two numerical experiments. Throughout we use $a = 0$, $b = 1$, which one may think of as a finite mass-spring chain with unit coupling and a single on-site stiffness defect of strength α at position k . In solid-state physics the same matrix appears as a tight-binding Hamiltonian for an electron hopping on a one-dimensional lattice with a single impurity site; the defect eigenvalue corresponds to a localized electronic state, and the envelope describes how quickly its probability amplitude decays away from the impurity.

- **Fig. 1 (defect eigenvalue vs. α).** We plot $\lambda(\alpha, k)$ as a function of α for a bulk site ($N = 200, k = 100$) and a near-edge site ($k = 10$). The dotted curves show the finite- N band edges $E_{\min}^{(N)}$ and $E_{\max}^{(N)}$. The black dashed line is the analytic upper bound $U_{\text{bound}}(\alpha)$ from Theorem 2. The defect eigenvalue curve lies strictly between the band edge and this bound, and its monotone increase with α is exactly the behavior guaranteed by Theorem 1.
- **Fig. 2 (semi-log localization and envelope).** For $N = 200$, $k = 100$, and $\alpha = 2$ we choose α so that $\lambda(\alpha, k) > a + 2|b|$ lies outside the band. On a semi-log scale we plot $|v_j^{(\alpha,k)}|$ versus $|j - k|$ (dots) together with the geometric envelope $|C r_-|^{|j-k|}$ from Theorem 3, where r_- is the small root of (1). The near-linear decay of the dots and their close adherence to the straight-line envelope confirm the predicted geometric localization; slight deviations near the ends of the chain reflect finite-size boundary effects.

The first experiment emphasizes how a single diagonal defect pulls one eigenvalue smoothly out of the band and how elementary matrix bounds already give practical enclosures for that eigenvalue. The second experiment shows that the associated defect mode is strongly localized, a feature that is crucial in applications such as designing optical waveguide arrays that trap light at a single channel or engineering impurity states that store energy in a localized vibration mode.

Omitted figures (exceed scope). Asymptotic-shift plots, edge-distance sweeps, and run-time/tolerance scaling offers other interesting relationships to study; they would further connect the spectral picture to physics of transport and localization in disordered chains.

V. Methods

We evaluate the resolvent entry

$$m_k(\lambda) = \frac{p_{k-1}(\lambda) p_{N-k}(\lambda)}{p_N(\lambda)}$$

using the three-term recurrence for $p_n(\lambda)$ together with floating-point scaling and Clenshaw/Gautschi-style stable evaluation of recurrences [9, 8]. For each trial λ we sweep the recurrence once from $n = 0$ up to $n = N$, normalizing as needed to avoid overflow and underflow; this gives an $O(N)$ evaluation of $m_k(\lambda)$ suitable for repeated use in the scalar function $f(\lambda) = 1 + \alpha m_k(\lambda)$. In concrete terms, this amounts to a single forward pass through the recurrence with occasional rescaling, followed by a simple formula that reads off the discrete Green's function at site k .

For $\alpha > 0$, Theorem 2 provides a finite- N upper bound $U_{\text{bound}}(\alpha)$ and a natural left endpoint just above the band edge $E_{\text{max}}^{(N)}$. We choose $\varepsilon \in [10^{-12}, 10^{-6}]$ so that $f(E_{\text{max}}^{(N)} + \varepsilon) < 0$ and apply monotone bisection on $[E_{\text{max}}^{(N)} + \varepsilon, U_{\text{bound}}(\alpha)]$ until the interval length falls below a prescribed tolerance. We deliberately avoid infinite-chain hyperbolic (cosh) or arccos parametrizations near the edge, so that the computation remains tied to the finite matrix and to the rigorous bracket.

For the localization envelope, we diagonalize $H(\alpha, k)$ with a standard symmetric eigensolver to obtain the normalized defect mode $v^{(\alpha, k)}$ and its eigenvalue $\lambda(\alpha, k)$. We then evaluate the small root r_- of the quadratic $br^2 + (a - \lambda)r + b = 0$ and apply Theorem 3 only when $\lambda(\alpha, k) \notin [a - 2|b|, a + 2|b|]$. In physical terms this procedure computes the normal mode pinned at the defect and compares its numerically observed decay with the theoretical geometric envelope, which in applications corresponds to how quickly energy or probability density leaks away from the impurity site.

VI. Discussion and limitations

The present analysis is restricted to a single on-site defect in a real-symmetric tridiagonal Toeplitz chain with constant off-diagonal coupling b . We do not treat complex hopping amplitudes, random coefficients, or interactions between multiple defects; in a tight-binding or mass-spring interpretation this corresponds to a perfectly ordered background with one modified site and no disorder. All bounds are proved at finite N and are uniform in the defect position k ; we do not analyze subtle issues in the thermodynamic limit $N \rightarrow \infty$ or edge scaling regimes.

Near the band edges we consciously avoid infinite-chain closed forms and cosh parametrizations, which can obscure the distinction between the finite- N spectrum and its limiting band. Instead we rely on the finite-chain band edges $E_{\text{min}}^{(N)}, E_{\text{max}}^{(N)}$, the scalar secular equation $f(\lambda) = 1 + \alpha m_k(\lambda)$, and the bracket from Theorem 2.

From a physics point of view, these results describe an idealized impurity problem in one dimension; phenomena such as disorder-induced localization, impurity bands, or higher-dimensional transport lie beyond the scope of the present finite-chain model, but they are natural directions for further study.

VII. Extended topics and further directions for "fun"

Edge-sensitive lifts. A refined lower (and upper) bound can incorporate the explicit distance-to-boundary parameter $d = \min\{k - 1, N - k\}$ through the finite- N Green's function factor (see [5, 6]). This would sharpen the bracket in Theorem 2 near the edges and explain more quantitatively how the defect level for a near-edge site lifts off the band differently from a bulk site, as suggested by Fig. 1.

Asymptotics in $|\alpha|$. Using a resolvent expansion around aI and the bound $\|T_N(a, b) - aI\| \leq 2|b|$, one can show that, for large $|\alpha|$,

$$\lambda(\alpha, k) = a + \alpha + \frac{b^2}{\alpha} + O(\alpha^{-3}) \quad (\alpha \rightarrow \pm\infty),$$

uniformly in N and k . A log-log plot of $\lambda(\alpha, k) - (a + \alpha)$ versus $1/\alpha$ illustrates the α^{-1} behavior. This refines Theorem 1 by describing how the defect eigenvalue peels away from the trivial shift $a + \alpha$ when the defect is very strong.

Complexity and certification. Viewing one evaluation of $m_k(\lambda)$ via the recurrence as $O(N)$ work, the bisection scheme in the Methods section has overall cost $O(N \log((U - L)/\varepsilon))$ to achieve accuracy ε on an initial interval $[L, U]$. A bisection certificate follows from the strict positivity of

$$f'(\lambda) = \alpha e_k^\top (T_N(a, b) - \lambda I)^{-2} e_k > 0,$$

which is another way to express the monotonicity already used in Theorem 1. This connects the scalar

defect-eigenvalue problem to standard numerical linear algebra notions of complexity and certified root-finding.

Higher dimensions and non-selfadjoint variants. On a higher-dimensional lattice, the Toeplitz structure is replaced by transfer matrices or block Jacobi operators, and Combes–Thomas–type bounds still guarantee exponential decay of localized modes but require sectorial resolvents and more functional analysis [7]. Non-selfadjoint variants with complex hopping amplitudes or gain/loss would link the present framework to models of optical waveguide arrays and non-Hermitian quantum systems, but fall outside the scope of the present note.

Any of these directions could be developed into a more extensive study, building directly on the elementary resolvent, rank-one perturbation, and localization results established here.

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Acknowledgment of Tools Used

Calculations and solution steps were verified using *Wolfram Alpha*, *ChatGPT (OpenAI)*, and *Mathway*. These resources were employed as computational aids and to cross-check intermediate steps for accuracy.

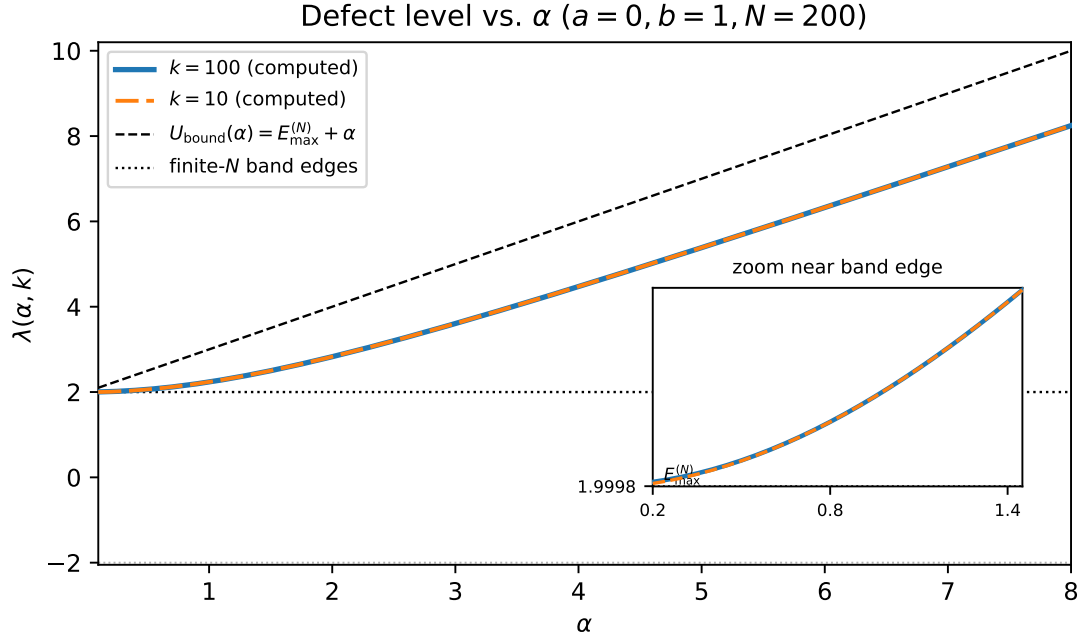


Figure 1: Defect level vs. α for $a = 0$, $b = 1$, $N = 200$. Blue solid: computed defect eigenvalue $\lambda(\alpha, k)$ for bulk $k = 100$. Orange dashed (slightly thinner): near-edge $k = 10$ (nearly overlaps the bulk curve). Black dashed: analytic upper bound $U_{\text{bound}}(\alpha)$ from Theorem 2. Dotted: finite- N band edges $E_{\min}^{(N)}, E_{\max}^{(N)}$; the monotone growth of the defect level with α illustrates Theorem 1.

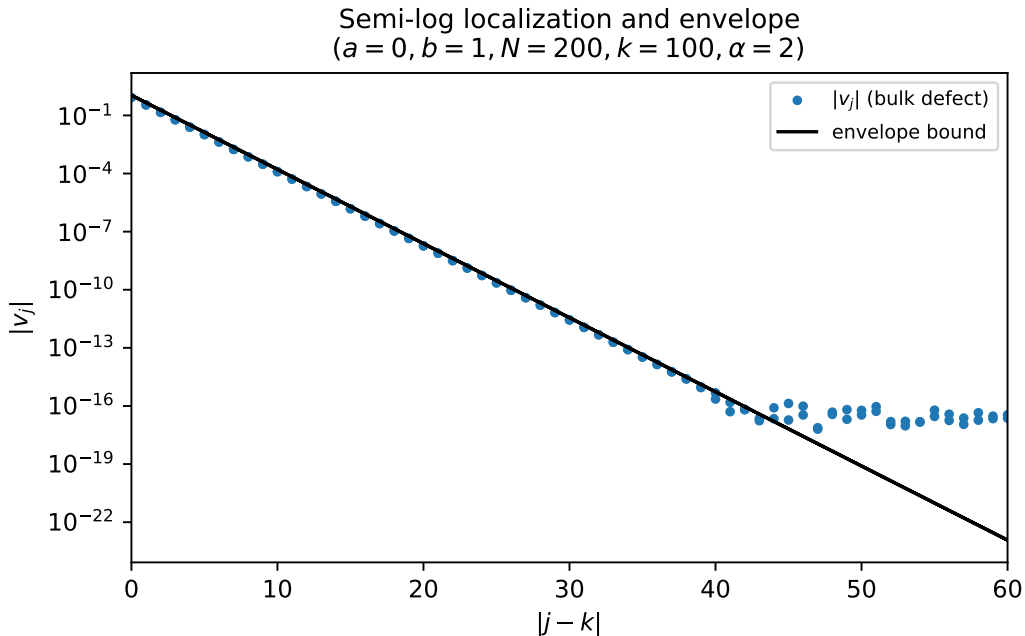


Figure 2: Semi-log localization and envelope for $a = 0$, $b = 1$, $N = 200$, $k = 100$, $\alpha = 2$ with $\lambda(\alpha, k) > a + 2|b|$. Dots: components $|v_j^{(\alpha, k)}|$ versus $|j - k|$. Solid black: envelope from Theorem 3, $C|r_-|^{d|j-k|}$ with r_- solving (1) and $d = \min\{k - 1, N - k\}$. The near-linearity of the dots on this semi-log plot confirms the geometric decay predicted analytically.